Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, SQL, Java, C/C++
Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, Feature Engineering, Exploratory Data Analysis, Supervised & Unsupervised Learning, Model Evaluation, Statistical Analysis, Time-Series Analysis, Anomaly Detection
Deep Learning & AI: PyTorch, TensorFlow/Keras, LSTM, Neural Networks, Generative AI, LLMs, RAG
Data Analytics & Visualization: Power BI, Tableau, Matplotlib, Plotly
Databases: DBMS, MySQL, SQLite, MongoDB, Redis
Backend & Tools: FastAPI, Flask, REST APIs, Git/GitHub, Docker, Streamlit
Engineering Practices: Data Preprocessing, Model Validation, PyTest, Test-Driven Development, Reproducible Experimentation

EXPERIENCE

AI for Sustainability Virtual Intern* |LINK** — ***July – Sep, 2026

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

***Docwise* |LINK**

- Built an AI-powered PDF question-answering system using Google Gemini and RAG, enabling contextual question answering over documents exceeding 200 pages.
- Developed a document processing and retrieval pipeline combining PDF extraction, chunking, embeddings, and semantic retrieval, achieving 95%+ query accuracy.
- Integrated FastAPI and LLM inference to deliver an end-to-end AI application for document understanding and conversational querying.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Engineered 15+ time-series features from 500K+ NSE market observations and trained Isolation Forest and LSTM Autoencoder models to detect anomalous price and volume movements.
- Applied FinBERT sentiment analysis to 25K+ financial news records and integrated sentiment scores with market features, combining structured market data and unstructured financial text into a unified analytical pipeline.
- Benchmarked 2 anomaly-detection models using precision, recall, F1-score, and inference time, with the best model targeting ~86% precision, ~81% recall, and ~83% F1-score on the evaluation dataset.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built an OCR-based data extraction pipeline using EasyOCR, OpenCV, and NLP to transform unstructured receipts into structured transaction data, improving processing efficiency by 40%.
- Developed data preprocessing and expense categorization workflows to clean, validate, and organize extracted transaction data for downstream analysis.
- Built analytics and reporting features for spend trends and budget tracking, converting transaction data into actionable financial insights for end users.

***CV-FL: Privacy-Preserving Federated Learning* | LINK**

- Designed a privacy-preserving federated learning framework for edge IoT networks, combining ternary sign quantization and super-increasing sequence compression to reduce communication overhead by ~19.7×.
- Implemented SignScore-based robust aggregation with resilience against up to 30% malicious clients, improving robustness against model poisoning attacks.
- Integrated Merkle-tree-based verification and dual-layer masking for lightweight model-update verification, achieving 100% tamper detection at ≥5% corruption.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |LINK — 2026
- AWS APAC Solutions Architecture virtual experience program | LINK — 2025